

```
In [1]: #list
l=[]
l
```

Out[1]: []

```
In [3]: l=[1,2,3,4,5]
l
```

Out[3]: [1, 2, 3, 4, 5]

```
In [4]: l.append(6)
l
```

Out[4]: [1, 2, 3, 4, 5, 6]

```
In [5]: l.append(7,8)
l
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[5], line 1
----> 1 l.append(7,8)
      2 l

TypeError: list.append() takes exactly one argument (2 given)
```

```
In [6]: l.append([7,8,9,0])
l
```

Out[6]: [1, 2, 3, 4, 5, 6, [7, 8, 9, 0]]

```
In [8]: l1=[22,45,'honey',1+2j,True]
l1
```

Out[8]: [22, 45, 'honey', (1+2j), True]

```
In [9]: print(len(l))
print(len(l1))
```

7
5

```
In [10]: #list membership
10 in l1
```

Out[10]: False

```
In [11]: l2=l1.copy()
l2
```

Out[11]: [22, 45, 'honey', (1+2j), True]

```
In [12]: print(id(l1))
print(id(l2))
```

1937447609216
1937441572864

```
In [14]: l2.clear()  
l2
```

```
Out[14]: []
```

```
In [16]: len(l2)
```

```
Out[16]: 0
```

```
In [17]: del l2
```

```
In [18]: l2
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[18], line 1  
----> 1 l2  
  
NameError: name 'l2' is not defined
```

```
l1.count(1)
```

```
In [19]: l1.count(1)
```

```
Out[19]: 1
```

```
In [20]: l[:]
```

```
Out[20]: [1, 2, 3, 4, 5, 6, [7, 8, 9, 0]]
```

```
In [21]: l[-1]
```

```
Out[21]: [7, 8, 9, 0]
```

```
In [22]: l1[100]
```

```
-----  
IndexError                                Traceback (most recent call last)  
Cell In[22], line 1  
----> 1 l1[100]  
  
IndexError: list index out of range
```

```
In [23]: l[3:-1]
```

```
Out[23]: [4, 5, 6]
```

```
In [24]: l[0:10:3]
```

```
Out[24]: [1, 4, [7, 8, 9, 0]]
```

```
In [25]: l[::1]
```

```
Out[25]: [1, 2, 3, 4, 5, 6, [7, 8, 9, 0]]
```

```
In [26]: l[::2]
```

```
Out[26]: [1, 3, 5, [7, 8, 9, 0]]
```

```
In [27]: l[::-1]
```

```
Out[27]: [[7, 8, 9, 0], 6, 5, 4, 3, 2, 1]
```

```
In [28]: l[:: -2]
```

```
Out[28]: [[7, 8, 9, 0], 5, 3, 1]
```

```
In [29]: l[0]
```

```
Out[29]: 1
```

```
In [31]: l[0]=100  
1
```

```
Out[31]: [100, 2, 3, 4, 5, 6, [7, 8, 9, 0]]
```

```
In [32]: l[-1]='honey singh'  
1
```

```
Out[32]: [100, 2, 3, 4, 5, 6, 'honey singh']
```

```
In [34]: l.extend(l1)  
1
```

```
Out[34]: [100,  
2,  
3,  
4,  
5,  
6,  
'honey singh',  
22,  
45,  
'honey',  
(1+2j),  
True,  
22,  
45,  
'honey',  
(1+2j),  
True]
```

```
In [35]: l.index('honey')
```

```
Out[35]: 9
```

```
In [37]: l.insert(2,25)  
1
```

```
Out[37]: [100,
          2,
          25,
          25,
          3,
          4,
          5,
          6,
          'honey singh',
          22,
          45,
          'honey',
          (1+2j),
          True,
          22,
          45,
          'honey',
          (1+2j),
          True]
```

```
In [40]: l.pop()
         1
         l.sort()
         1
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[40], line 3
      1 l.pop()
      2 1
----> 3 l.sort()
      4 1

TypeError: '<' not supported between instances of 'str' and 'int'
```

```
In [41]: 1
```

```
Out[41]: [2,
          3,
          4,
          5,
          6,
          25,
          25,
          100,
          'honey singh',
          22,
          45,
          'honey',
          (1+2j),
          True,
          22,
          45]
```

```
In [46]: l=list(map(str,l))
         l.sort()
         print(l)
```

```
['(1+2j)', '100', '2', '22', '22', '25', '25', '3', '4', '45', '45', '5', '6', 'True', 'honey', 'honey singh']
```

```
In [48]: l.sort(key=str)
1
```

```
Out[48]: ['(1+2j)',
'100',
'2',
'22',
'22',
'25',
'25',
'3',
'4',
'45',
'45',
'5',
'6',
'True',
'honey',
'honey singh']
```

```
In [49]: l=list(map(int,l))
l.sort()
print(l)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[49], line 1
----> 1 l=list(map(int,l))
      2 l.sort()
      3 print(l)

ValueError: invalid literal for int() with base 10: '(1+2j)'
```

```
In [51]: l.sort(reverse=False)
1
```

```
Out[51]: ['(1+2j)',
'100',
'2',
'22',
'22',
'25',
'25',
'3',
'4',
'45',
'45',
'5',
'6',
'True',
'honey',
'honey singh']
```

```
In [52]: 1
```

```
Out[52]: ['(1+2j)',  
          '100',  
          '2',  
          '22',  
          '22',  
          '25',  
          '25',  
          '3',  
          '4',  
          '45',  
          '45',  
          '5',  
          '6',  
          'True',  
          'honey',  
          'honey singh']
```

```
In [58]: l[15][7]
```

```
Out[58]: 'i'
```

```
In [59]: for i in l:  
          print(i)
```

```
(1+2j)  
100  
2  
22  
22  
25  
25  
3  
4  
45  
45  
5  
6  
True  
honey  
honey singh
```

```
In [60]: for i in enumerate(l):  
          print(i)
```

```
(0, '(1+2j)')  
(1, '100')  
(2, '2')  
(3, '22')  
(4, '22')  
(5, '25')  
(6, '25')  
(7, '3')  
(8, '4')  
(9, '45')  
(10, '45')  
(11, '5')  
(12, '6')  
(13, 'True')  
(14, 'honey')  
(15, 'honey singh')
```

```
In [61]: all(1)
```

```
Out[61]: True
```

```
In [62]: any(1)
```

```
Out[62]: True
```